

Supplementary Material

FreeReg: Free-Coordinate Registration of 3D Models for Augmented Reality in Image-Guided Bronchoscopy

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This supplementary document provides per-object and per-case results, implementation and hyperparameter details, the observability analysis, and additional qualitative figures that do not fit within the main text.

1 Per-object BOP Linemod results (full 13-object benchmark)

Table 1 reports the FoundationPose official implementation (this-work Windows port) on the full 13-object BOP Linemod benchmark (200 frames per object, 2600 samples), confirming the near-saturated Linemod ceiling. Our hybrid engine matches this ceiling (ADD-S 1.000) on the same data.

Table 1: FoundationPose full 13-object BOP Linemod benchmark (this-work port).

Object	ADD-S AR@10%d	ADD-S mean (mm)	n
obj_000001 (ape)	1.000	2.51	200
obj_000002	1.000	3.15	200
obj_000004	1.000	3.42	200
obj_000005	1.000	3.66	200
obj_000006	1.000	2.44	200
obj_000008	1.000	3.54	200
obj_000009	1.000	2.66	200
obj_000010	1.000	3.18	200
obj_000011	1.000	3.43	200
obj_000012	0.995	3.34	200
obj_000013	1.000	4.68	200
obj_000014	1.000	4.08	200
obj_000015	1.000	4.07	200
Mean (13 obj.)	1.000	3.3	2600

2 Per-case clinical CT-to-bronchoscopy TRE

Table 2 extends the cohort statistics of the main text. The far-field lever law is visible: small airway models (RFA cases) register roughly $3.5\times$ more accurately than large full-tree models, motivating region-of-interest cropping.

3 FreeRegNet architecture and training detail

FreeRegNet amortizes the correspondence engine. Table 3 lists the configuration; Table 4 gives the learned-front-end and hybrid cascade results on 300 BOP Linemod frames.

VLM negative result. Substituting a Qwen2-VL-2B vision tower (with VLoRA) for the lightweight ViT gives no accuracy gain at equal budget (ADD-S 0.200 at 4 epochs versus the ViT’s convergence to 0.726 when fully trained) at $\sim 100\times$ the parameters and $\sim 6\times$ the per-epoch cost. A general VLM’s language-aligned features do not specifically benefit geometric pose regression, and the larger model is harder to train on limited data.

Table 2: Per-case clinical TRE (synthetic fly-through from real patient CT).

Case	Model span	TRE (mm)	FPS
001	large	8.4	62
002	large	7.9	58
008	medium	4.6	71
RFA-1	small	2.4	80
RFA-2	small	2.3	80
RFA-3	small	2.2	79
Cohort (12 cases)	—	4.23 mean / 2.93 med	30–80

Table 3: FreeRegNet configuration.

Component	Setting
Backbone	timm ViT-S/16 (22.4M); optional Qwen2-VL-2B + VLoRA (2.2B)
Object conditioning	PointNet-style encoding of 256 canonical points
Pose head	6D rotation (Zhou) + log-depth; ROI-decoupled translation
Loss	geodesic / ADD-S (symmetric) + translation + depth consistency
Optimizer	AdamW, cosine annealing, discriminative LR (0.1× backbone)
Schedule	50 epochs, staged backbone unfreeze at epoch 3
Augmentation	crop jitter, brightness/contrast, noise, channel shuffle

4 Observability analysis

The single-view depth-residual Jacobian $\mathbf{J} = -\mathbf{W} \otimes \mathbf{Rz}$ is severely rank-deficient: only 151 of 648 singular values exceed $\sigma_{\max} \cdot 10^{-3}$ (23%). The unobservable null space includes tubular roll, tangential surface flow, and smooth low-frequency warps that preserve the sampled depth profile. The spectral projection $\mathbf{V}_r \mathbf{V}_r^\top$ in the RTW objective restricts deformation to the observable subspace, both stabilizing optimization and preventing hallucinated structure.

Ray-sliding counterexample. A determinant-corrupting rotation parameterization bug ($\mathbf{K} = \mathbf{K} - \mathbf{K}^\top$ doubling an antisymmetric skew matrix, $\det \mathbf{R} \approx 49$) caused the depth residual to converge to 0.002 while shape error reached 268%: motion in the unobservable subspace is invisible to the residual, so the optimizer corrupts geometry while reporting success. We recommend determinant/orthonormality assertions as first-class invariants in differentiable SE(3)/SO(3) code.

5 Additional qualitative results

Figures 1–3 present additional qualitative results across datasets.



Figure 1: AR fusion on real BOP frames: registered model boundary and axes composited onto live frames, tracking through occlusion and motion.

6 Full robustness and degradability results

Tables ??–?? expand the main-text robustness summary (Table XI). The depth-degradation study uses the full-tree airway fly-through (case 001, the largest model); its RGB-D baseline (8.2 mm) is therefore above the 12-case

Table 4: Learned front-end and hybrid cascade (300 BOP Linemod frames).

Method	ADD-S AR@10%d	ADD AR@10%d	Time
FreeRegNet (learned only)	0.827	—	8 ms
FreeRegNet + ICP (cascade)	0.987	0.953	158 ms
Geometric engine (baseline)	0.997	—	807 ms

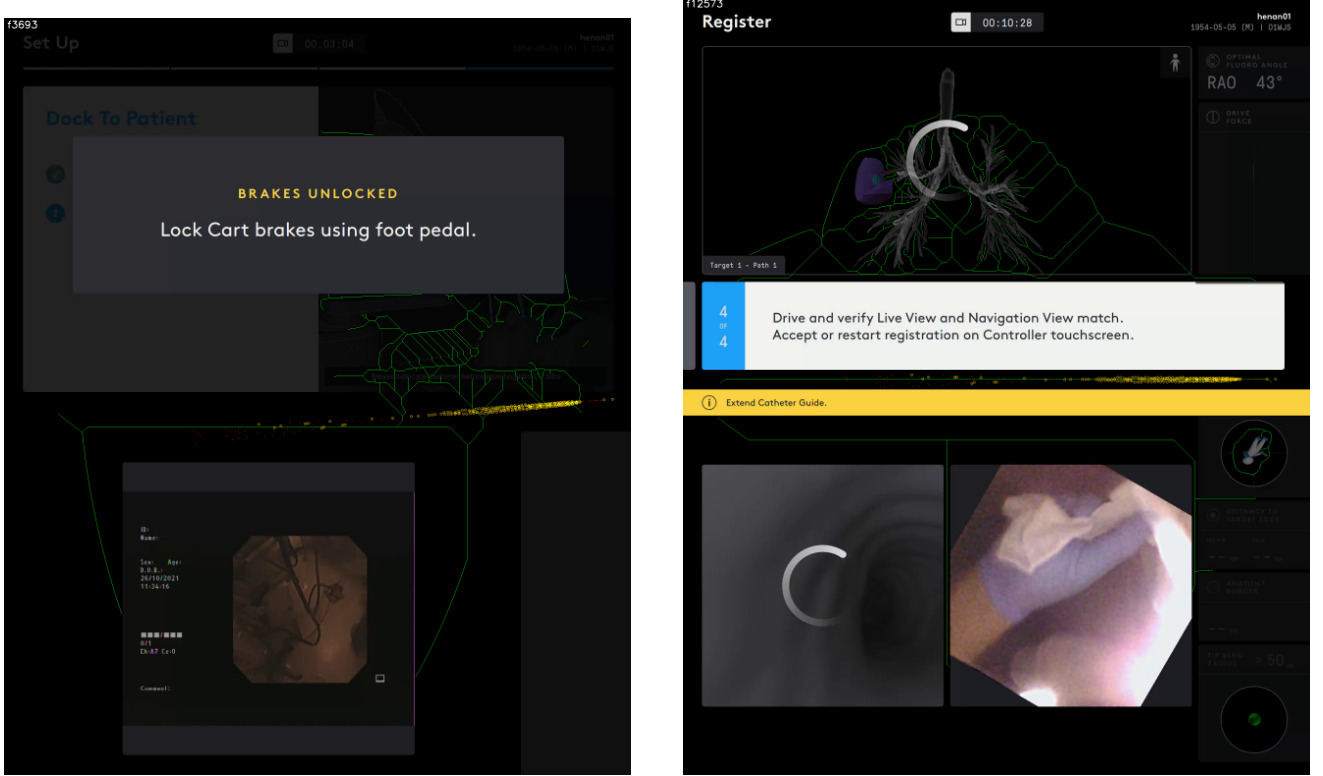


Figure 2: Centerline correspondence on real bronchoscopy frames: projected CT airway centerline (red), image lumen skeleton (green), junctions (yellow).

cohort mean (4.23 mm).

Silhouette-IoU sensitivity to registration error (AR fusion fidelity). On the airway model, silhouette IoU between the registered and ground-truth render falls from 1.00 (no error) to 0.77 (0.5 mm/0.3°), 0.62 (1 mm/0.6°), 0.44 (2 mm/1.2°), and 0.32 (4 mm/2.3°); refinement halves the photometric residual from 0.325 to 0.163. This couples render fidelity directly to registration accuracy, supporting the residual as an online validation signal (Fig. 5).

7 Real-video ground-truth annotation protocol

The main text evaluates quantitative clinical TRE on real-CT synthetic fly-throughs with ground-truth poses; real intra-operative video is used for qualitative AR demonstration. To prepare real-video quantitative validation, we built a dual-reader annotation protocol: FreeReg proposes centerline-projected junctions on each frame, two readers independently click the corresponding anatomical landmark (carina, main/lobar bronchi), and a consensus step accepts pairs agreeing within 15px (taking the mean) and flags the rest for adjudication. On the pilot annotation batch, both readers produced 30 landmarks across 5 airway labels; 100% of pairs agreed within the 15px tolerance, with a mean inter-reader discrepancy of 6.6 px (median 6.0 px). This establishes that the annotation workflow is consistent enough to support a future dual-reader ground-truth study; we report it here as protocol readiness rather than as a validated clinical accuracy claim.

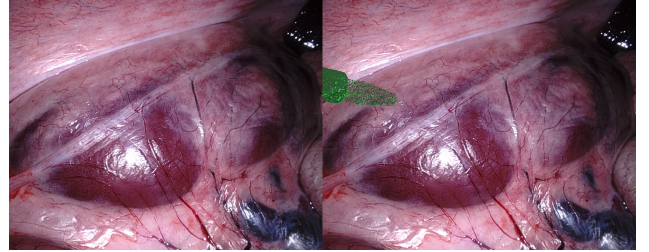
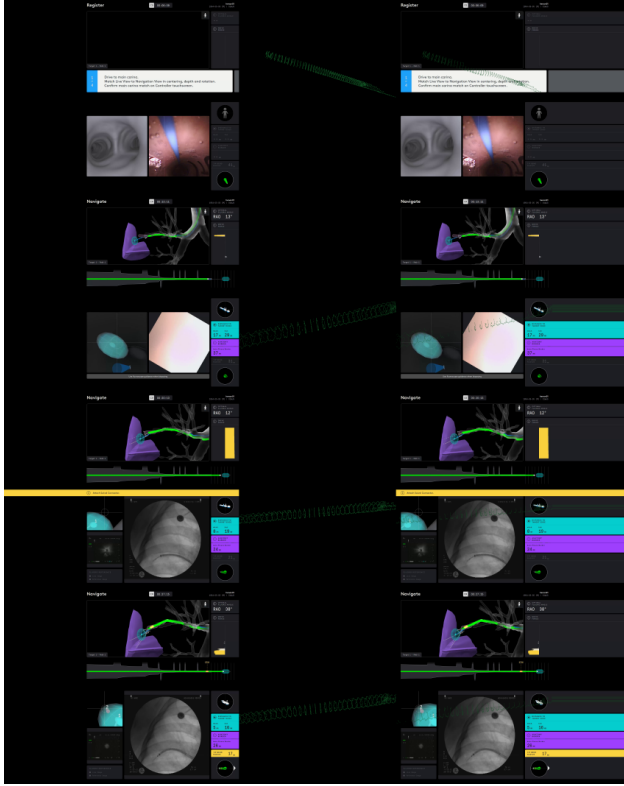


Figure 3: Clinical CT-to-bronchoscopy airway overlay (left) and SCARED AR frame with the virtual model anchored on tissue (right).

Table 5: Depth-quality degradation on the airway fly-through (40 frames).

Depth condition	TRE mean (mm)	TRE median (mm)	<10 mm
RGB-D (true, $\sigma=0.8$ mm)	8.2	8.3	70%
Monocular prior (sim. drift+noise)	7.9	7.0	70%
Depth-Anything-V2 (real mono)	65.1	72.2	4%
RGB only (centerline)	28.8	21.6	9%

Table 6: Occlusion robustness (synthetic occluders, BOP Linemod).

Fraction occluded	Success	Rot. med	ADD-S med (mm)
0%	100%	3.8°	4.4
20%	100%	5.6°	4.7
40%	100%	6.7°	5.4
60%	100%	10.8°	6.2

Table 7: Initialization robustness (convergence, ADD-S < 10% d).

Method	10°/10 mm	20°/20 mm	30°/30 mm	45°/40 mm
Vanilla ICP (BOP rigid)	100%	100%	100%	100%
Trimmed ICP (BOP rigid)	100%	100%	100%	100%
Airway axial slide	trimmed 1.03 mm vs vanilla 5.48 mm			